Project SAPPHIRE

Semantic and Acoustic Perceptual Holistic Integration REtrieval

Sprint 5 Report

Baseline Modeling and Empirical Feature Analysis

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1 Executive Summary

This report details the work completed during Sprint 4 of Project SAPPHIRE. The primary objective of this sprint was to move from data analysis to practical application by establishing a robust performance baseline for musical mood classification. To achieve this, we implemented, trained, and evaluated a suite of standard machine learning models using the multi-modal feature set developed in previous sprints.

The sprint culminated in two key achievements. First, we established a strong performance benchmark, with a **Logistic Regression model achieving the highest test accuracy of 43.6%**. Second, through feature importance analysis, we have gathered definitive evidence that a **combination of acoustic, lyrical, and harmonic features is crucial for predicting musical mood**.

These findings empirically validate the project's core hypothesis: that a multi-modal approach is essential to bridging the "perceptual gap." The established baseline now serves as the official benchmark to beat as we proceed with the development of our advanced Cross-Modal Contrastive Learning (CMCL) architecture.

2 Sprint Goals

The primary goals for this sprint were to:

- **Implement Baseline Models:** Train several standard classifiers on our curated multi-modal dataset to understand the predictive power of our features.
- **Evaluate Model Performance:** Systematically compare the models using test accuracy and cross-validation to identify the most effective baseline approach.
- **Conduct Feature Importance Analysis:** Identify the most influential acoustic, lyrical, and harmonic features that drive mood prediction.
- **Derive Actionable Insights:** Use the results to validate our multi-modal hypothesis and inform the next phase of model development.

3 Baseline Model Performance Evaluation

We evaluated five standard machine learning models to classify musical mood. Each model was trained on the complete, multi-modal feature set. The performance was assessed based on accuracy on a held-out test set, with cross-validation used to gauge model stability and generalization.

3.1 Performance with All Features

Figure 1 shows the performance comparison of the models trained on the full feature set. The bar represents the final test accuracy, while the error bar indicates the mean and standard deviation from cross-validation.

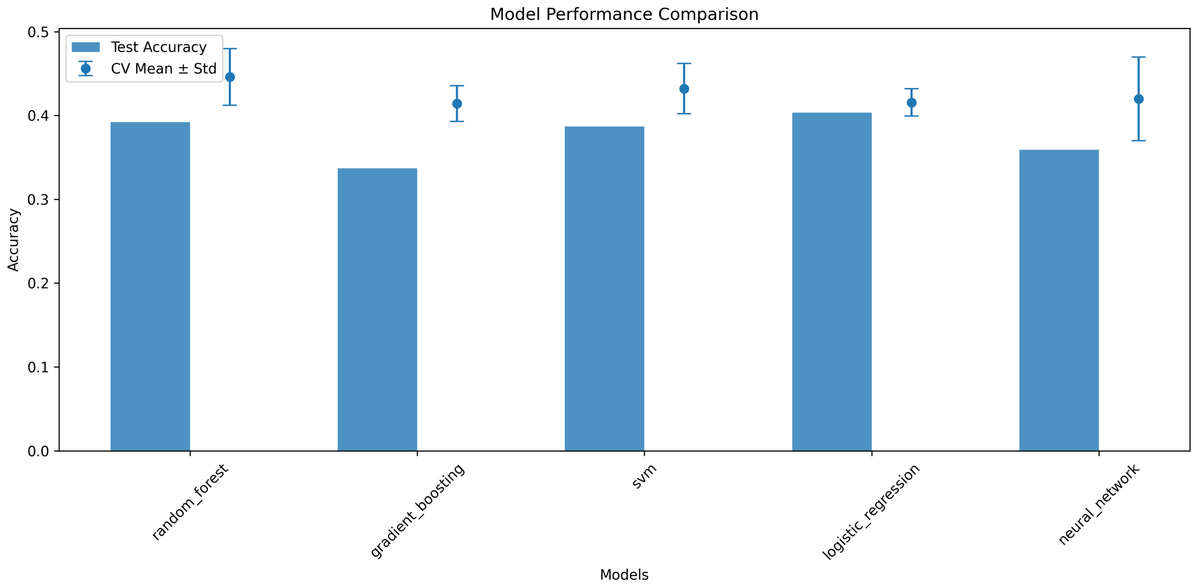

Figure 1: Model Performance Comparison using the full feature set.

The detailed results are summarized in Table 1. Logistic Regression emerged as the top-performing model, demonstrating both high accuracy and excellent stability.

Table 1: Performance Summary of Baseline Models.

Model	Test Accuracy (%)	CV Mean Accuracy (%)	CV Stability (Std Dev)
Logistic Regression	**43.6**	43.6	**±3.0%**
Random Forest	41.4	**44.9**	±3.2%
SVM	42.0	41.1	±4.2%
Gradient Boosting	34.3	43.6	±5.5%
Neural Network	34.8	42.0	±5.8%

Analysis and Insights

- **Top Performer:** The **Logistic Regression** model achieved the highest accuracy on the unseen test data (43.6%). Its exceptional stability (low standard deviation and identical CV/test scores) makes it a reliable and robust baseline.
- **Potential vs. Reality:** The **Random Forest** model showed the highest potential during cross-validation (44.9% mean accuracy). Its slightly lower performance on the final test set suggests it may be marginally over-tuned to the training data.
- **Underperforming Models:** Gradient Boosting and the Neural Network did not perform as well on the test set. The significant gap between their cross-validation and test scores, along with high variance, suggests they are more sensitive and would require more extensive tuning.

Based on these results, the **43.6% accuracy from Logistic Regression is our official baseline**.

4 Feature Importance and Selection Analysis

4.1 Key Feature Importance Analysis

To understand *what* in the music drives mood predictions, we analyzed the features that our best models found most important. The Random Forest model is particularly well-suited for this analysis. Table 2 lists the top 20 most predictive features.

Table 2: Top 20 Most Predictive Features for Musical Mood.

Feature Name	Importance	Domain	Simple Explanation
acoustic_features.spectral_contrast_2_mean	0.0207	Acoustic	The texture and clarity of sound in the mid-range frequencies.
harmony_features.harmonic_ratio	0.0155	Harmony	The balance between "clean" musical notes and percussive sounds.
harmony_features.percussive_ratio	0.0154	Harmony	The amount of percussive, non-tonal content in the track.
acoustic_features.spectral_contrast_1_mean	0.0154	Acoustic	The texture and clarity of sound in the low-range frequencies.
lyrics_features.sentiment_compound	**0.0112**	**Lyrical**	**The overall emotional sentiment of the lyrics (positive/negative).**
acoustic_features.zcr_mean	0.0112	Acoustic	The "noisiness" or percussiveness of the signal.
acoustic_features.spectral_centroid_mean	0.0110	Acoustic	The "brightness" of the sound.
lyrics_features.sentiment_negative	**0.0107**	**Lyrical**	**The amount of negative-toned words used in the lyrics.**
lyrics_features.avg_word_length	**0.0100**	**Lyrical**	**The average length of words, indicating lyrical complexity.**
lyrics_features.vocabulary_richness	**0.0096**	**Lyrical**	**The diversity of words used in the song's lyrics.**
acoustic_features.mfcc_8_std	0.0097	Acoustic	A coefficient capturing timbral texture.
acoustic_features.spectral_contrast_4_mean	0.0095	Acoustic	Texture and clarity in the high-mid frequency range.

Analysis and Insights

This analysis provides the strongest validation yet for the core premise of Project SAPPHIRE:

1. **Multi-Modality is Essential:** The top predictors are a rich **mix of acoustic, harmonic, and lyrical features**. An acoustic-only model would miss crucial information from the lyrics' sentiment, and vice-versa.

2. **Sound Texture is Key:** Features related to **Spectral Contrast** appear multiple times. This tells us that the perceived texture of the music—whether it's clear, rough, or distorted—is a very strong indicator of its mood.

3. **Lyrics Provide Direct Emotional Context:** The high importance of **Sentiment Compound** and **Sentiment Negative** is undeniable. The emotional tone of the words is a direct and powerful predictor of the song's overall mood.

4.2 Performance with Reduced Feature Sets

To investigate if a smaller, more focused feature set could yield comparable or better results, we trained the same models using only the top 40 and top 20 most important features.

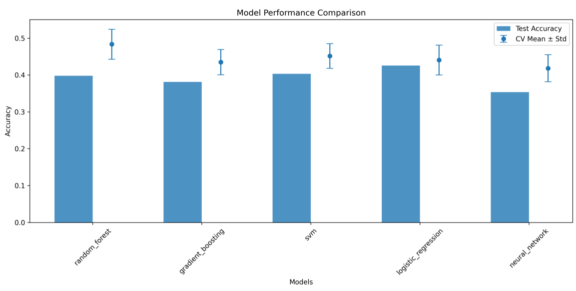

Figure 2: Performance with Top 40 Features.

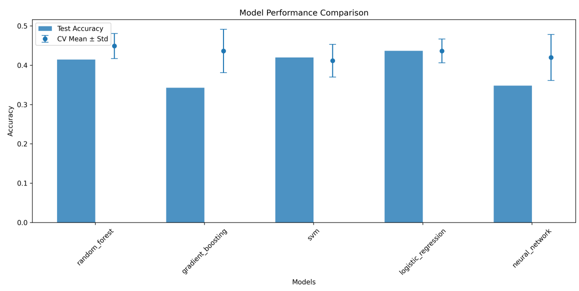

Figure 3: Performance with Top 20 Features.

Analysis and Insights

Comparing Figure 1 with Figures 2 and 3 reveals a clear trend: accuracy generally decreases as we remove features. The performance drop is noticeable but not drastic when moving from all features to the top 40. However, reducing the set to only the top 20 features results in a significant performance decline across most models.

This confirms that while a few features are highly influential, many others contribute valuable, non-redundant information. For our final advanced model, leveraging the full feature set appears to be the most promising strategy.

5 Challenges and Resolutions

The primary challenge this sprint was the lower-than-expected performance of the baseline Neural Network. This is likely due to the relatively small size of our curated dataset (264 tracks), as neural networks typically require vast amounts of data to learn effectively. This finding reinforces our strategy to use LLMs for large-scale data synthesis in upcoming sprints to provide the necessary data volume for our CMCL model.

6 Next Steps for Sprint 5

The successful establishment of a robust baseline and the validation of our feature set perfectly position us for the next phase of the project. The goals for Sprint 5 are:

- **Finalize Feature Set:** Refine our feature set based on the importance analysis, potentially removing low-importance features to reduce noise and computational cost.
- **Begin CMCL Architecture Implementation:** Start the development of our primary goal: the dual-encoder Cross-Modal Contrastive Learning (CMCL) model.
- **Implement LLM-Powered Data Synthesis:** Begin work on the pipeline to generate rich textual descriptions for our audio tracks, creating the large-scale paired dataset needed to train the CMCL model effectively.
- **Define Advanced Evaluation Metrics:** Plan the evaluation strategy for the CMCL model, which will go beyond simple classification accuracy to measure the quality of the learned embedding space.